

PARTS LIST

Semiconductors:

IC1	- LM7812, 12V regulator
IC2	- TLC555 timer
T1	- 2N2219 NPN transistor
D1	- 1N4007 rectifier diode
D2	- 1N4148 signal diode
BR1	- 1A bridge rectifier
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R3	- 1k Ω
R4	- 580k Ω
R5	- 1.2M Ω
R6	- 1.8M Ω
R7	- 2.2M Ω
R8	- 3.3M Ω
R9	- 680 Ω

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 470 μ F, 35V electrolytic
C3	- 10nF, ceramic disc

Miscellaneous:

CON1-CON3	- 2-pin connector
RL1	- 12V SPDT relay
S1, S2	- Push-to-on switch
S3-S7	- On/off switch
X1	- 230V AC primary to 12V AV, 500mA secondary transformer
	- Wires and jumper wires, as needed
	- 230V AC appliance
	- 230V AC power cord

panel of the cabinet and connected to the PCB using external wires.

Alternatively, the components can be assembled on a general-purpose PCB if a custom-designed PCB is not available. It is vital to ensure proper heat dissipation for the LM7812 voltage regulator. A small heat-sink should be attached to the regulator to prevent overheating.

Before powering

the circuit for the first time, all connections should be carefully verified and double-checked to ensure accuracy. The polarities of the electrolytic capacitors, diodes (D1 and D2), and the transistor (T1) must be checked. Ensuring correct orientation and secure connections is essential to prevent component damage and to ensure reliable operation.

To test the circuit, a 230V AC appliance, such as a lamp or fan, should be connected. The desired delay time can be selected by pressing one of the selector switches (S3 through S7). Then, the start switch (S1) should be pressed to trigger the monostable timer circuit. This energises the relay and powers the connected appliance. The appliance will remain on for the selected delay duration. Once the preset time has elapsed, the TLC555 timer output goes low, de-energising the relay and automatically turning off the appliance.

If required, the stop switch (S2) can be pressed at any time to manually interrupt the operation and turn off the appliance before the timer completes its cycle. **EFY**

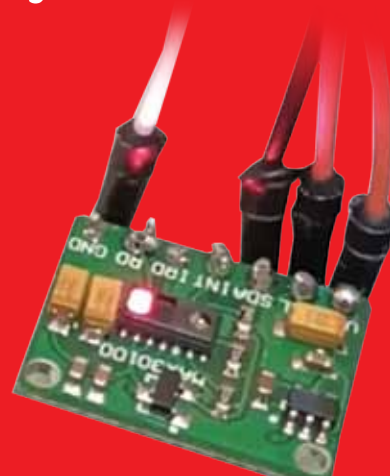
Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-delay-off-appliance-timer>

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